

Canonical Liftings

Beyond Signature Compression

From a work by:

Alessandro Budroni, Giuseppe D'Alconzo, Leonardo Errati, Edoardo Persichetti

Politecnico di Torino
Department of Mathematical Sciences

Group Actions I

Let $(G, \circ)$ be a group and X a set. A *group action of G on X* is a map

$$\star : G \times X \rightarrow X$$

compatible with the group operation:

$$\text{id}_G \star x = x$$

$$g \star (h \star x) = (g \circ h) \star x$$

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X = possible states of the cube

$$G = (Z_3^7 \times Z_2^{11}) \rtimes ((A_8 \times A_{12}) \rtimes Z_2).$$

Group Actions II

Sigma Protocol

$$\begin{array}{ccc} x & \xrightarrow{g} & g \star x \\ & \searrow \rho \circ g & \downarrow \rho \\ & & \rho \star (g \star x) \end{array}$$

GAIP

Given $x, y \in X$, find $g \in G$ with $x = g \star y$.

Canonical Liftings I

A *canonical lifting* for a group action $(G, X, \star)$ is a tuple $(R, C, \text{Rep}, \text{Can}, \text{Lift})$ of sets and efficient maps such that this diagram commutes:

$$\begin{array}{ccc} G \times X & \xrightarrow{(\text{Rep}, \text{id})} & R \times X \\ \star \downarrow & & \downarrow \text{Lift} \\ X & \xrightarrow{\text{Can}} & C \end{array}$$

$$\begin{aligned} \text{Rep} &: G \rightarrow R \\ \text{Can} &: X \rightarrow C \\ \text{Lift} &: R \times X \rightarrow C \end{aligned}$$

Defining Property

$$\text{Can}(g \star x) = \text{Lift}(\text{Rep}(g), x)$$

Canonical Liftings II

A canonical lifting is *effective* if

- 1 it is efficient to find a g with given representative r
- 2 if $\text{Can}(x) = \text{Can}(y)$, it is efficient to find g with $y = g \star x$.

CLCP

Given $x, y \in X$, find $r, s \in R$ with $\text{Lift}(r, x) = \text{Lift}(s, y)$.

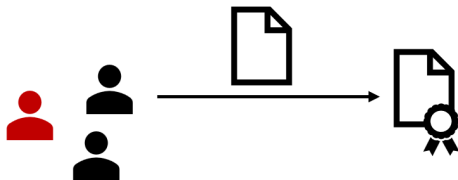
If a canonical lifting is *effective*, GAIP and CLCP are PPT-equivalent.

Ring Signatures [RST01]

ring of public keys $\mathcal{R} = (pk_1, \dots, pk_n)$

A *ring signature* allows an user to sign on behalf of $\mathcal{R}$ without revealing its identity.

RS = (KeyGen, Sign, Verify)

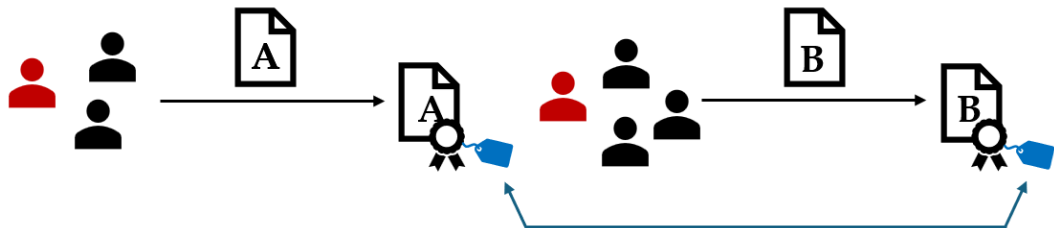


Linkable Ring Signatures [LWW04]

ring of public keys $\mathcal{R} = (pk_1, \dots, pk_n)$

Linkable if it enables to verify whether two signatures were produced by the same sk_i .

LRS = (KeyGen, Sign, Verify, *Link*)



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Security properties:

Anonymity, hide which ring member signed.

Linkability, detect repeated use of the same signing key.

Non-frameability, prevent signatures falsely linked to an honest user.

Advanced Signatures via Canonical Liftings

Signature: Fiat–Shamir on the sigma protocol. Security problem: CLCP.

RS: Fiat–Shamir on the OR sigma protocol. Security problem: CLCP.

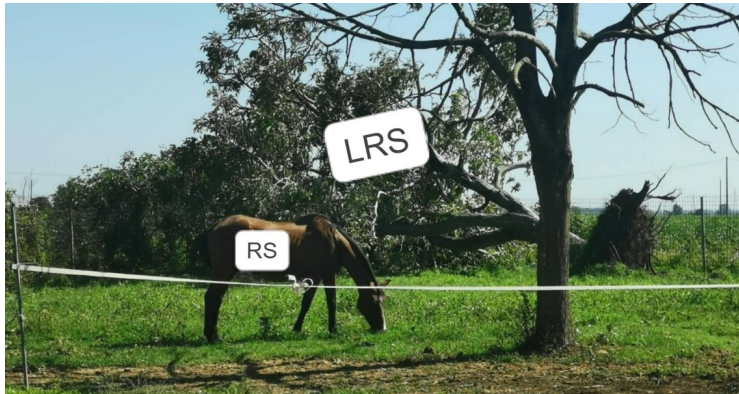
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LRS from Group Actions [BKP20]

RS: for a commutative group action, apply Fiat–Shamir to a sigma protocol for

$$\mathcal{R}^{\text{OR}} = \left\{ \langle (x_1, \dots, x_N), w \rangle \in X^N \times W \mid \text{for some } i \in [N], (x_i, w) \in \mathcal{R} \right\}.$$

LRS: tag-key relation via witness-reuse.

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LRS: tag-key relation via witness-reuse.

With LEP, witness-reuse leads to key recovery attacks [BCDD⁺24].

$$\text{pk} = \mathbf{Q}\mathbf{G}, \quad \text{tg} = \mathbf{Q}^{-1}\mathbf{G}$$

Non-commutative actions may not meet the security hypotheses of LRS [IGL26].

LRS from Canonical Liftings

Idea: tag-key relation is g, g' related by $\text{Rep}(g) = \text{Rep}(g')$.

Left-compatibility

- 1 $\text{Rep}(g) = \text{Rep}(h) \implies \text{Rep}(\rho \circ g) = \text{Rep}(\rho \circ h)$
- 2 given g , sampling uniform h such that $\text{Rep}(g) = \text{Rep}(h)$ is efficient

linkability

non-splitting
+ left-compatible

linkable anonymity

2-pseudorandom
+ left-compatible

non-frameability

frame one-way
+ left-compatible

LRS from Canonical Liftings

The case of LEP

LEP: $Q = PD, Q' = PD'$ related by P

$$\text{pk}_i = Q_i G, \quad \text{tg}_i = Q'_i T$$

Left-compatibility

- 1 if Q, Q' share permutation, so do MQ, MQ'
- 2 sampling uniform Q'' with same permutation is efficient

linkability

if $\text{Aut}(G), \text{Aut}(T) = \{1\}$,
it holds unconditionally

linkable anonymity

best known attacks are
those on LEP

non-frameability

best known attacks are
those on LEP

Performance of CELeReS

Scheme	Security	2^1	2^2	2^6	2^{10}	2^{12}	Assumption
CELeReS [BDEP26]	128 bit	26.7	27.8	32.3	36.7	39.0	OW, F-OW, 2-PR
	192 bit	69.2	72.3	85.1	97.8	104.2	
	256 bit	119.1	123.8	142.6	161.3	170.7	
Raptor [LAZ19]	100 bit	–	–	82.7	1301.9	5203.3	NTRU
Calamari [BKP20]	NIST-I	3.6	4.6	14.1	20.1	23.1	CSIDH-512
Falafl [BKP20]	NIST-I	33	34	36	39.3	41	MSIS, MLWE
DualRing-PRF [ZSL ⁺ 24]	128 bit	–	–	45	64	–	Residue PRFs

Scheme	Ring size	KeyGen	Sign	Verify
CELeReS	2^1	$1.32\text{ M} / 4.5 \times 10^{-4}\text{ s}$	$421\text{ M} / 0.141\text{ s}$	$397\text{ M} / 0.137\text{ s}$
	2^3		$2419\text{ M} / 0.806\text{ s}$	$2035\text{ M} / 0.678\text{ s}$
	2^6		$9300\text{ M} / 3.090\text{ s}$	$7865\text{ M} / 2.631\text{ s}$

CELeReS-o (I)

Variant with self-orthogonal tag matrices.

Security	2^1	2^2	2^6	2^{10}	2^{12}
128 bit	18.9	19.9	24.5	28.9	31.2
192 bit	50.8	54.0	66.7	79.5	85.9
256 bit	85.4	90.1	108.8	127.6	136.9

Ring size	KeyGen	Sign	Verify
2^1		511 M/0.171 s	477 M/0.159 s
2^3	55.0 M/0.018 s	2508 M/0.834 s	2090 M/0.697 s
2^6		9836 M/3.255 s	7647 M/2.543 s

CELeReS-o (II)



Summary

What we showed

- Canonical liftings abstract compression techniques
- They can build classical and (linkable) ring signatures

Open problems

- Different group actions
- New techniques from liftings
- For code-based LRS: tag compression techniques

<https://eprint.iacr.org/2026/1348>

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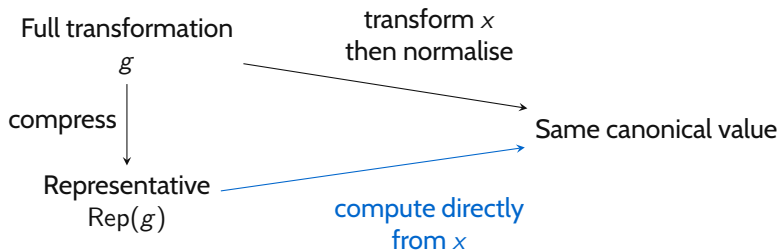
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EXTRA SLIDES

open in case of emergency

Canonical Liftings Ia



Defining Property

$$\text{Can}(g \star x) = \text{Lift}(\text{Rep}(g), x)$$

Which lifting for CELeReS?

The canonical form used in LESS is not left-compatible.
Instead, we adapt existing constructions.

$$X = \mathbb{F}_q^{k \times n}, \quad G = \text{Mon}_n(\mathbb{F}_q)$$

$$C = \text{CF}(X) \text{ [DKQ}^+25], \quad R = \text{Perm}_n(\mathbb{F}_q)$$

$$\begin{array}{ccc} G \times X & \xrightarrow{(\text{Rep}, \text{id})} & R \times X \\ \star \downarrow & & \downarrow \text{Lift} \\ X & \xrightarrow{\text{Can}} & C \end{array}$$

$$\begin{aligned} \text{Rep} &: \mathbf{PD} \mapsto \mathbf{P} \\ \text{Can} &: \mathbf{M} \mapsto \text{CF}(\mathbf{M}) \\ \text{Lift} &: (\mathbf{M}, \mathbf{P}) \mapsto \text{CF}(\mathbf{MP}) \end{aligned}$$

Security Properties of LRS

linkability

Find α in $\text{MAut}(\mathbf{G}) \setminus \text{MAut}(\mathbf{T})$.

If $\mathbf{G}, \mathbf{T}$ are random, they are trivial with overwhelming probability.

linkable anonymity

Distinguish between
 $(\text{RREF}(\mathbf{GQ}), \text{RREF}(\mathbf{TQ}'))$
 $(\text{RREF}(\mathbf{GQ}_0), \text{RREF}(\mathbf{GQ}_1))$

A 2-sample version of 2-pseudorandomness for the quotient action by $\text{Perm}_n(\mathbb{F}_q)$.
The best attacks are those on LEP.

non-frameability

(If $\text{MAut}(\mathbf{T})$ is trivial)
Given $(\mathbf{G}, \mathbf{G}')$ and $(\mathbf{T}, \mathbf{T}')$ pairs of linear equivalent codes under the same $\mathbf{P}$, find $\mathbf{P}$.

The best attacks are those on LEP.